

MLII ECG Anomaly Detection and Arrhythmia Classification

Corrected retrospective temporal evaluation

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Can a single-lead system detect abnormal beats and assign a supported subtype in later ECG from patients whose earlier ECG was available for training?

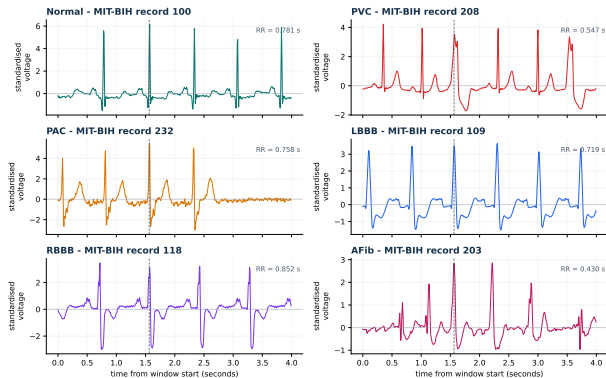
- Binary decision: N or abnormal.
- Conditional subtype: PVC, PAC, LBBB, RBBB or AFib.
- Targets: whole-system accuracy above 90%, lower 95% bound above 90%, interval width at most three percentage points.

Interpretation

The temporal test was observed during earlier development. This is a retrospective known-patient evaluation.

Why morphology and rhythm both matter

Real labeled ECG segments from the locked final 20%



Dashed line: annotated R peak used to align the four-second window.

Illustrative MITDB MLII examples. PVC and

bundle-branch patterns alter shape; premature beats and AF rhythms also require timing context.

Source: [1, 2]

The output codes are annotation groups

Code	Reference annotation	Interpretation	
N	N, e, j outside AFIB/AFL	Normal-like and escape group	Other supported beat
PVC	V, E	Ventricular premature/escape	
PAC	A, a, J, S	Supraventricular premature group	
LBBB	L	Left bundle-branch beat	
RBBB	R	Right bundle-branch beat	
AFib	N-like beat in AFIB/AFL	Fibrillation/flutter context	

symbols keep their morphology label during AFIB/AFL. Unsupported abnormalities enter binary testing only.

These codes are not pure clinical diagnoses or the standard AAMI five classes. AFib recall is not AF-only or episode-level sensitivity.

Source: [1]

Data sources and independent subjects

Source	Role
NSRDB	18 subjects; first stored ECG channel, ECG1. 15 subjects fit the autoencoder; 3 validate it.
MITDB	46 MLII records; 45 subject clusters. Earlier beats develop the hierarchy; later beats form the temporal test.
INCART	Complete cached I01 and I02 recordings, lead II. Both belong to source patient 1. Post-fit external check only.

NSRDB has no

significant arrhythmias at cohort level; individual windows were not screened using beat annotations.

Source: [3, 1, 4]

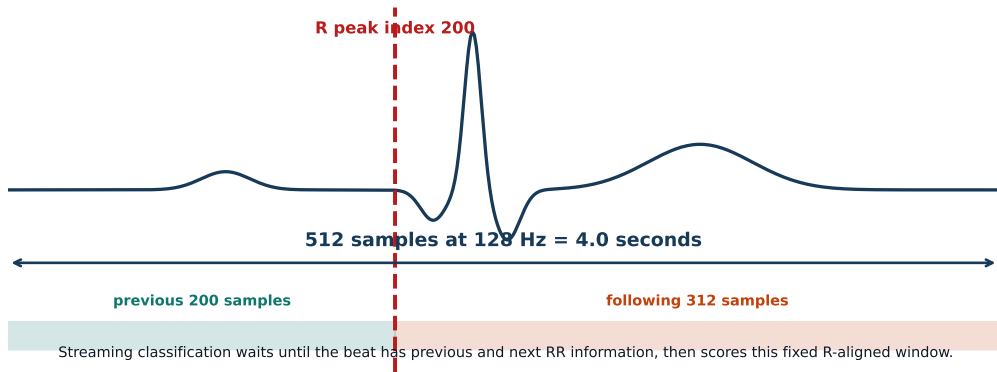
- NSRDB ECG1 supplies normal-reference reconstruction; its anatomical lead is unspecified.
- MITDB MLII is selected by header name. Record 114 stores it second.
- Records 102 and 104 lack MLII and are excluded.
- INCART standard lead II introduces a lead and acquisition shift.
- The software checks a sensor's declared lead, but cannot verify physical electrode placement.

Source: [3, 1, 4]

- ① Resample the one-channel signal to 128 Hz.
- ② Extract 512 samples around an R peak at index 200.
- ③ Filter within that four-second window: Butterworth 0.5–50 Hz, forward and backward.
- ④ Standardise that window to zero mean and unit variance.
- ⑤ Require finite, non-flat input and observed previous/next R peaks.

The same local preparation now supplies supervised fitting, offline scoring and streaming classification. This removes the original recording-wide normalisation and its test-region scale information.

Beat context creates a deliberate delay



A beat needs approximately 2.44 seconds of post-R signal and its next observed R peak. Chunk buffering and computation add delay. Offline experiments use annotations. Live detection errors and hardware latency require separate measurement.

- Dilated U-Net with residual convolutions and scaled skip connections.
- 2,680,641 trainable parameters; input and output shape 1×512 .
- 288,673 fitting windows and 57,177 validation windows.
- Two epochs; Adam learning rate 0.001; batch size 256.
- Denoising noise standard deviation 0.03; best validation MSE 0.061193.

The checkpoint remains frozen. Two epochs do not establish optimal convergence, and no ablation establishes the residual branch's independent benefit.

Source: [5, 6]

The two ensembles share 242 features

Feature group	Count
Whole-window mean bins	64
Central morphology bins	80
Central derivative bins	60
Relative frequency-band powers	8
Signal summary statistics	11
RR timing: 7 base plus 8 historical values	15
Autoencoder reconstruction residuals	4
Total	242

The four residuals are whole-window

MAE/MSE, central MAE and mean absolute residual change.

Let p_i be the ensemble abnormality score and q_{ic} the conditional subtype score for class $c \in \{1, \dots, 5\}$.

$$\hat{y}_i = \begin{cases} 0 \text{ (N)} & p_i \leq 0.5708333333, \\ \arg \max_{c \in \{1, \dots, 5\}} q_{ic} & p_i > 0.5708333333. \end{cases}$$

- Detector: 240 Extra Trees, square-root feature sampling.
- Subtype model: 260 Extra Trees, feature sampling 0.33.
- Both use class/subject weights and entropy splits.

Every gated beat receives a supported subtype. There is no validated unknown-class rejection or calibrated disease probability.

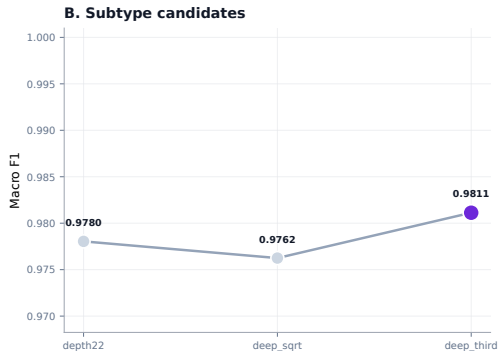
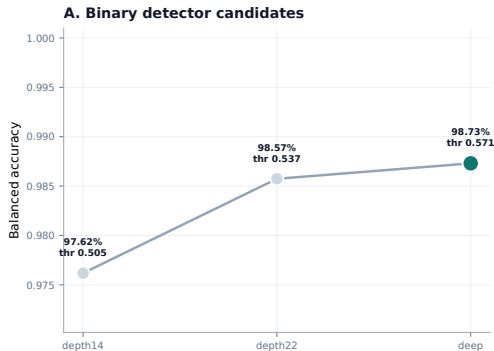
Source: [7]

Region	Corrected accepted beats
Candidate training: first 64%, less gap	66,285
Selection validation: 64–80%, less gap	16,029
Final refit: first 80%, less gap	83,050
Temporal test: final 20%	20,969

- Percentages count accepted beats, not exact elapsed recording time.
- Remove 16 beats before validation and test boundaries.
- Check sample separation, including a resampling margin.
- Test exposure in earlier development remains a source of optimism.

Candidate selection uses the development region

Validation-region model selection



Binary

candidates rank by balanced accuracy then F1. Subtype candidates rank by macro F1 then accuracy. The test does not choose the threshold in this rerun.

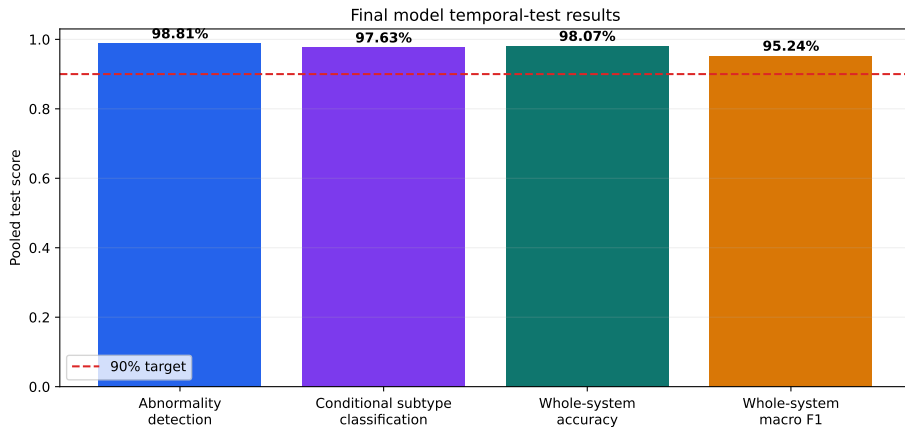
Three evaluations have different denominators

Evaluation	Beats	Meaning	There are 926
Binary	20,969	N versus any accepted abnormality	
Conditional subtype	7,958	True supported abnormalities, ignoring gate	
Whole system	20,043	N plus five supported groups, after gate	

unsupported abnormal test beats. They contribute to binary metrics but not to the six-class confusion matrix.

Whole-system accuracy includes correct N outputs and can remain high when a less common subtype has poor recall.

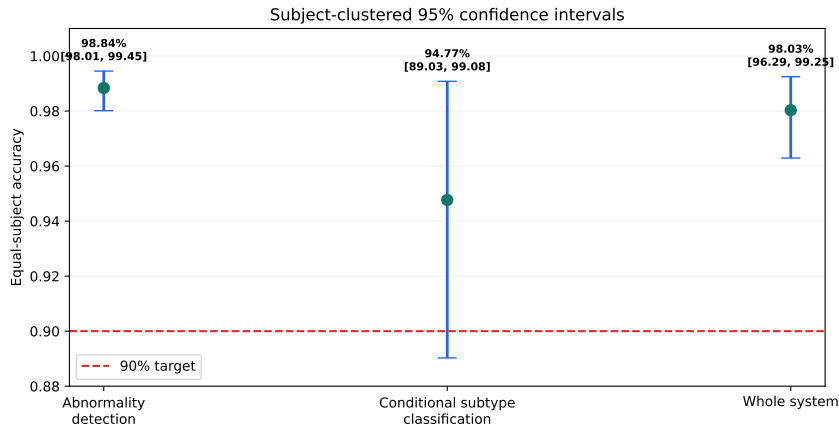
Corrected temporal whole-system accuracy is 98.07%



Binary accuracy

98.81%; conditional subtype accuracy 97.63%; whole-system macro F1 0.9524. Retrospective known-patient results.

Equal-subject accuracy and uncertainty



Whole-system subject mean

98.03%; 95% interval 96.29–99.25%; width 2.95 percentage points. The interval describes the subject mean, not pooled accuracy.

How the confidence interval was calculated

- 1 Combine MITDB records 201 and 202 as one subject.
- 2 Compute accuracy in each of the 45 subject clusters.
- 3 Resample 45 clusters with replacement, with equal subject weight.
- 4 Repeat 10,000 times; take the 2.5th and 97.5th percentiles.

The fitted model stays fixed. The interval does not include retraining variability, development-selection bias or a new population shift.

43 of 45 clusters exceed 90%. The sign test concerns the fraction of subjects above the target, not pooled beat accuracy.

Source: [8]

PAC recall remains below the target

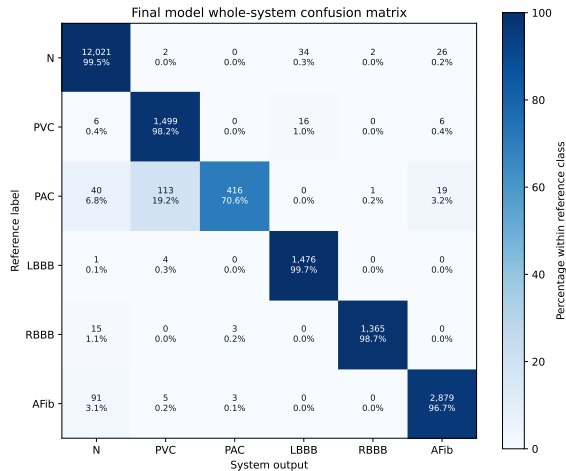
Group	True	Correct	Precision	Recall
PVC	1,527	1,499	92.36%	98.17%
PAC	589	416	98.58%	70.63%
LBBB	1,481	1,476	96.72%	99.66%
RBBB	1,383	1,365	99.78%	98.70%
AFib group	2,978	2,879	98.26%	96.68%

549 PAC beats pass the gate; 441 are correctly

named without gating; 416 are correct end to end.

These are annotation-group recalls, not patient diagnoses.

Whole-system errors



113 PAC beats become PVC. The gate sends 91 AFib-group beats and 40 PAC beats to N.

External check covers one INCART source patient

Fixed post-fit check	Result
I01 and I02; both patient 1	5,416 beats
Reference N / PVC	4,845 / 571
Binary accuracy / balanced accuracy	94.11% / 96.55%
Sensitivity / specificity	99.65% / 93.46%
Binary precision	64.22%
False alarms / missed abnormalities	317 / 2
Whole-system accuracy	92.85%
End-to-end PVC recall	87.74%

The threshold stays fixed. The subset has

no reference support for PAC, LBBB, RBBB or AFib and cannot establish population generalisation.

Source: [4]

What the prototype establishes

- A working single-lead hierarchy with reproducible preparation and scoring.
- Aggregate accuracy, lower confidence bound and interval-width targets met in the retrospective temporal run.
- Persistent PAC weakness and external false alarms.
- Consistent offline/streaming feature construction when peak locations agree.

What remains unvalidated

An untouched patient-disjoint test, all-class external recall, real AD8232 performance, clinical signal quality and prospective use.

- Earlier test exposure can bias design choices; the rerun cannot undo it.
- The NSRDB channel differs from MLII, and normality is a cohort assumption.
- The autoencoder's added value needs a controlled ablation.
- Labels combine beat morphology and rhythm context.
- A new study must reserve independent patients before any model selection.
- Sensor validation must measure peak detection, false alarms and total latency.

The corrected hierarchy reaches 98.07% whole-system accuracy on the retrospective MITDB temporal test, with an equal-subject 95% interval of 96.29–99.25%.

PAC recall remains 70.63%. The one-patient INCART check shows 92.85% whole-system accuracy and substantial false alarms.

The research contribution is a reproducible prototype with explicit evaluation limits. Clinical diagnostic performance has not been established.

Stage	Channels $\times$ samples	
Input	1×512	
Encoder blocks	32×256 , 64×128 , 128×64 , 256×32	The first up block
Dilated bottleneck	256×32 ; dilations 1, 2, 4, 8	
Decoder after skip resizing	128×32 , 64×64 , 32×128 , 32×256	
Linear output then interpolation	1×256 then 1×512	

resizes to its length-32 skip. A final interpolation restores the input length. The diagram and report describe the saved implementation, not an idealised doubling at every decoder stage.

With $e_t = x_t - \hat{x}_t$, the four residuals are

$$\frac{1}{512} \sum |e_t|, \quad \frac{1}{512} \sum e_t^2, \quad \frac{1}{240} \sum_{t=104}^{343} |e_t|, \quad \frac{1}{511} \sum_{t=1}^{511} |e_t - e_{t-1}|.$$

The seven base RR values describe previous/next interval, local mean, interval ratios, local variation and heart rate.

The eight historical values use up to 20 completed intervals: median, IQR, MAD, RMSSD, pNN50, relative range, trend and latest change. Only the next-RR base features require the next beat.

For class c and subject cluster s ,

$$w_{cs} \propto \frac{1}{n_c} \left(\frac{n_c}{S_c n_{cs}} \right)^\alpha.$$

- n_c : class support; S_c : subjects containing the class.
- n_{cs} : beats of class c in subject cluster s .
- Normalise weights to mean one and clip to $[0.05, 20]$.
- Selected α : 0.35 for binary, 0.25 for subtype.

Records 201 and 202 form one cluster during weighting and uncertainty analysis.

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